



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Class Activity – 20/07/2026

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1. Hospital Patient Admission Analysis

R Code:

```
# Install Packages (Run only once)
```

```
install.packages("ggplot2")
```

```
install.packages("treemap")
```

```
# Load Libraries
```

```
library(ggplot2)
```

```
library(treemap)
```

```
# Create Dataset
```

```
hospital <- data.frame(
```

```
  Department = c(
```

```
    rep("Cardiology",5),
```

```
    rep("Neurology",5),
```

```
    rep("Pediatrics",5),
```

```
    rep("Orthopedics",5),
```

```
    rep("Emergency",5)
```

```
),
```

```
  Admissions = c(
```

```
    120,135,140,128,150,
```

```
    90,95,105,100,110,
```

```
    180,170,190,200,210,
```

```
    130,125,145,138,150,
```

```
    240,250,260,255,270
```

```
)
```

```

)
#-----

# 1. Bar Chart
#-----

bar_data <- aggregate(Admissions~Department,hospital,sum)
ggplot(bar_data,
       aes(x=Department,y=Admissions,fill=Department))+
geom_bar(stat="identity")+
labs(title="Hospital Patient Admissions",
      x="Department",
      y="Total Admissions")+
theme_minimal()
#-----

# 2. Stacked Bar Chart
#-----

hospital$Quarter <- rep(c("Q1","Q2","Q3","Q4","Q1"),5)
ggplot(hospital,
       aes(x=Quarter,y=Admissions,fill=Department))+
geom_bar(stat="identity")+
labs(title="Department-wise Admissions by Quarter",
      x="Quarter",
      y="Admissions")+
theme_minimal()
#-----

# 3. Treemap
#-----

treemap(bar_data,
       index="Department",
       vSize="Admissions",
       title="Hospital Admission Treemap")

```

```
#-----
```

```
# 4. Box Plot
```

```
#-----
```

```
ggplot(hospital,  
  aes(x=Department,  
    y=Admissions,  
    fill=Department))+  
geom_boxplot()+  
labs(title="Admission Variability",  
  x="Department",  
  y="Admissions")+  
theme_minimal()
```

```
#-----
```

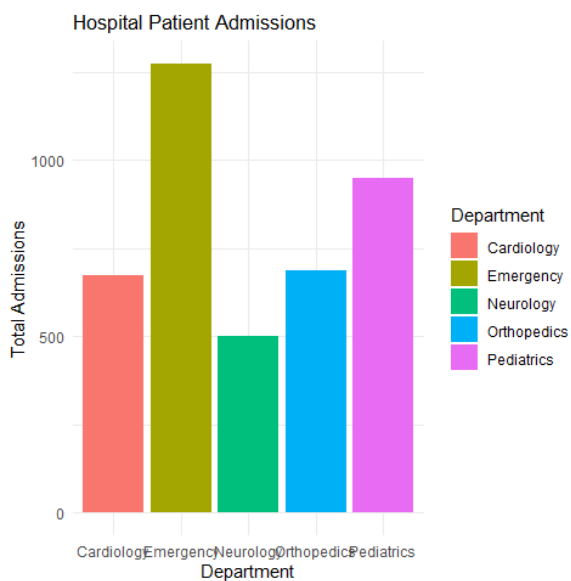
```
# 5. Histogram
```

```
#-----
```

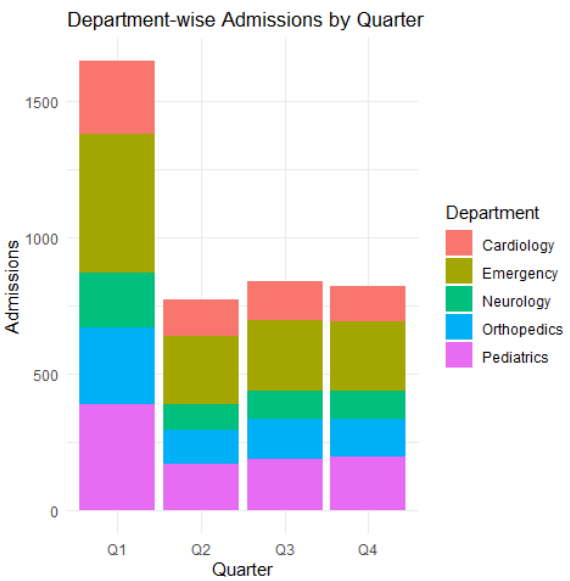
```
ggplot(hospital,  
  aes(x=Admissions,  
    fill=Department))+  
geom_histogram(binwidth=15,  
  color="black",  
  alpha=0.7)+  
labs(title="Distribution of Admissions",  
  x="Admissions",  
  y="Frequency")+  
theme_minimal()
```

Output:

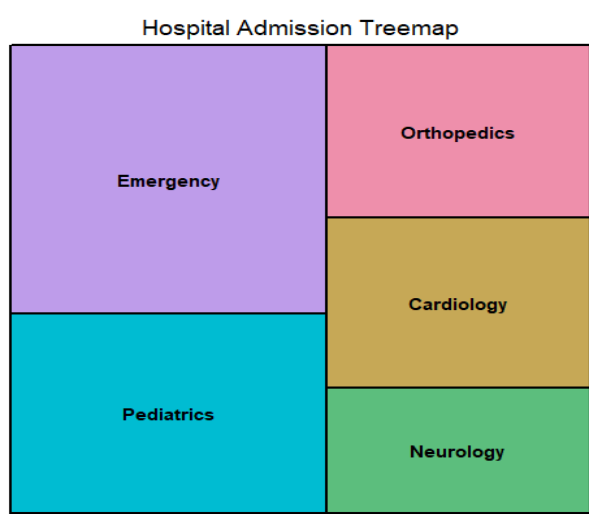
Bar Chart:



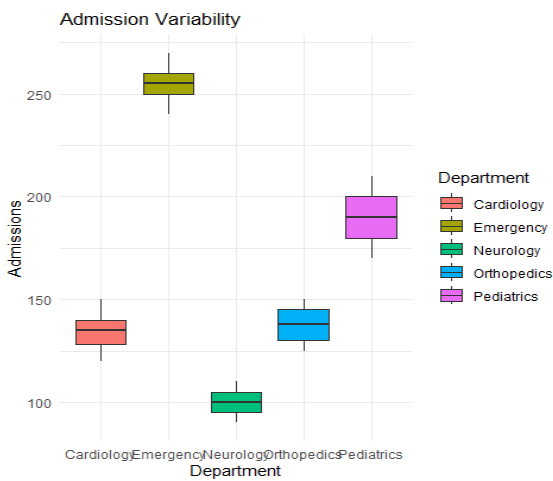
Stacked Bar Chart:



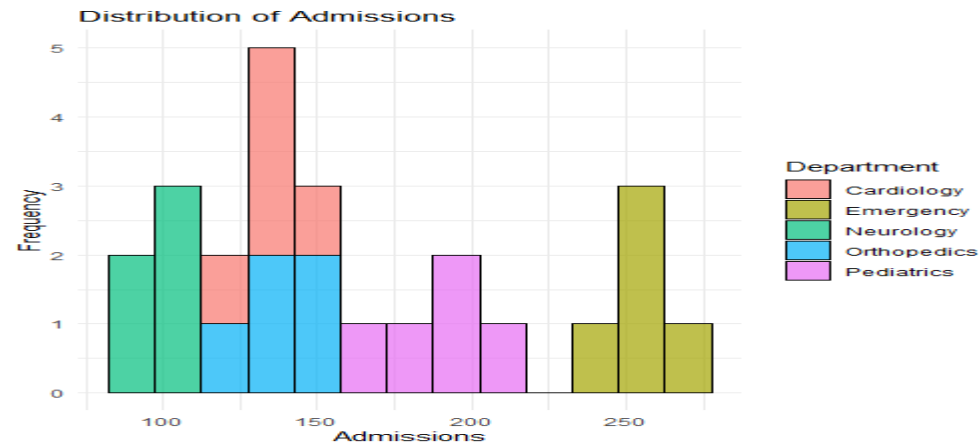
Treemap:



Box Plot:



Histogram:

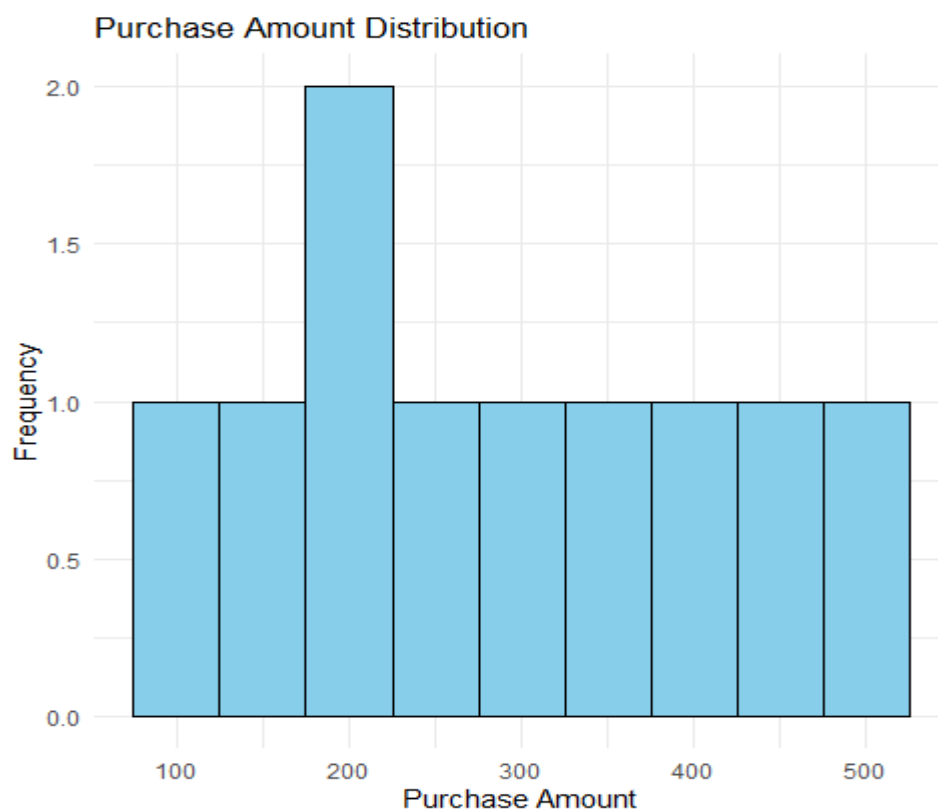


2. E-Commerce Customer Spending Behaviour.

Histogram:

```
ggplot(ecommerce,  
  aes(x = PurchaseAmount)) +  
geom_histogram(binwidth = 50,  
  fill = "skyblue",  
  color = "black") +  
labs(title = "Purchase Amount Distribution",  
  x = "Purchase Amount",  
  y = "Frequency") +  
theme_minimal()
```

Output:

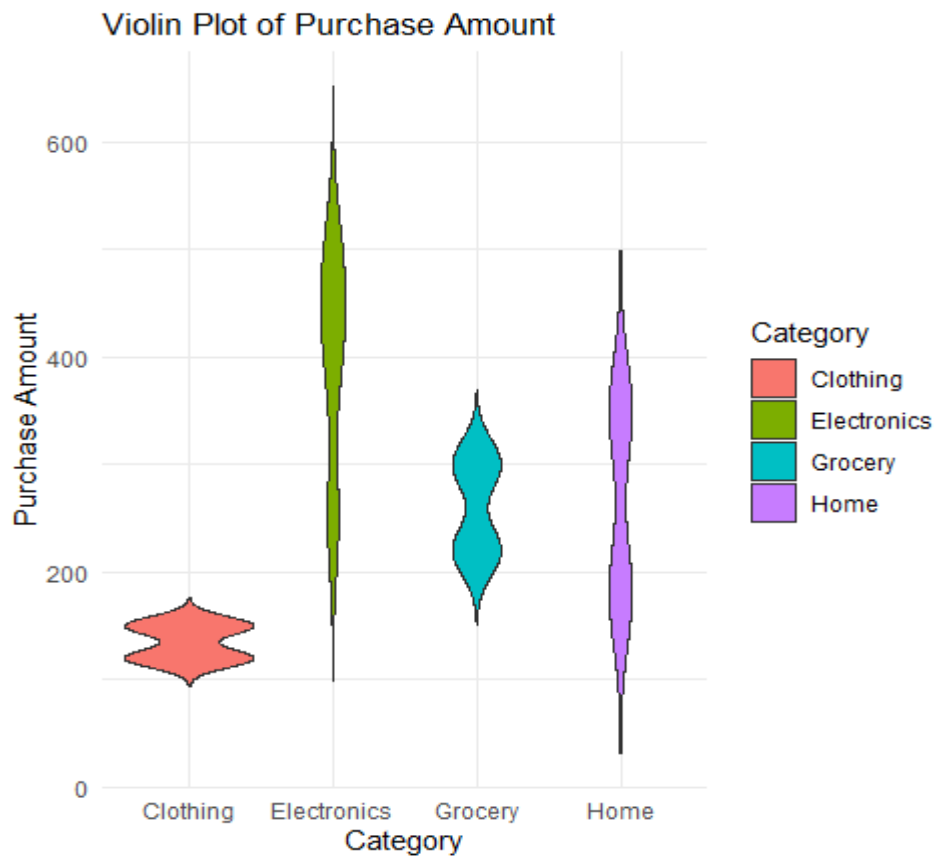


Violin Plot:

```
ggplot(ecommerce,  
  aes(x = Category,  
    y = PurchaseAmount,  
    fill = Category)) +
```

```
geom_violin(trim = FALSE) +
labs(title = "Violin Plot of Purchase Amount",
      x = "Category",
      y = "Purchase Amount") +
theme_minimal()
```

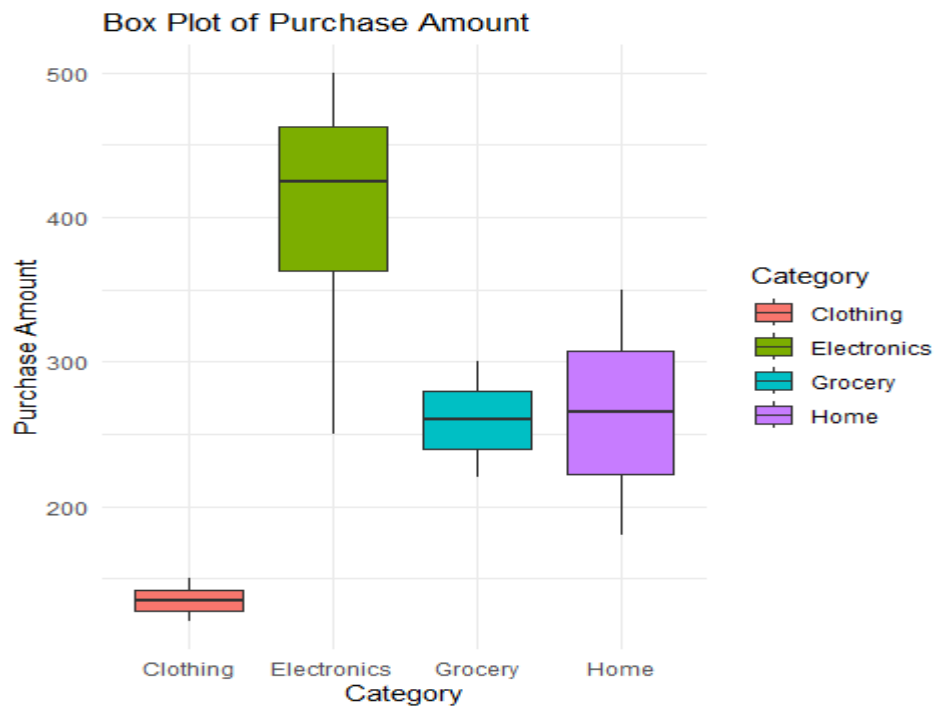
Output:



Box Plot

```
ggplot(ecommerce,
      aes(x = Category,
          y = PurchaseAmount,
          fill = Category)) +
geom_boxplot() +
labs(title = "Box Plot of Purchase Amount",
      x = "Category",
      y = "Purchase Amount") +
theme_minimal()
```

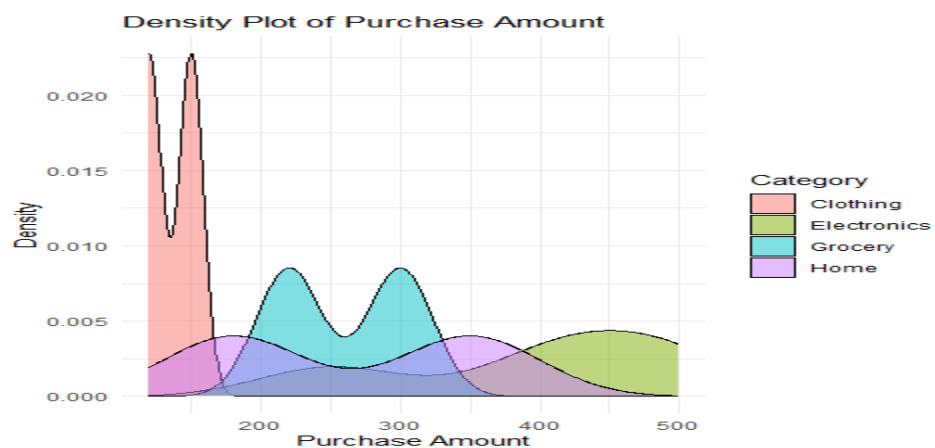
Output:



Density Plot:

```
ggplot(ecommerce,  
  aes(x = PurchaseAmount,  
    fill = Category)) +  
  geom_density(alpha = 0.5) +  
  labs(title = "Density Plot of Purchase Amount",  
    x = "Purchase Amount",  
    y = "Density") +  
  theme_minimal()
```

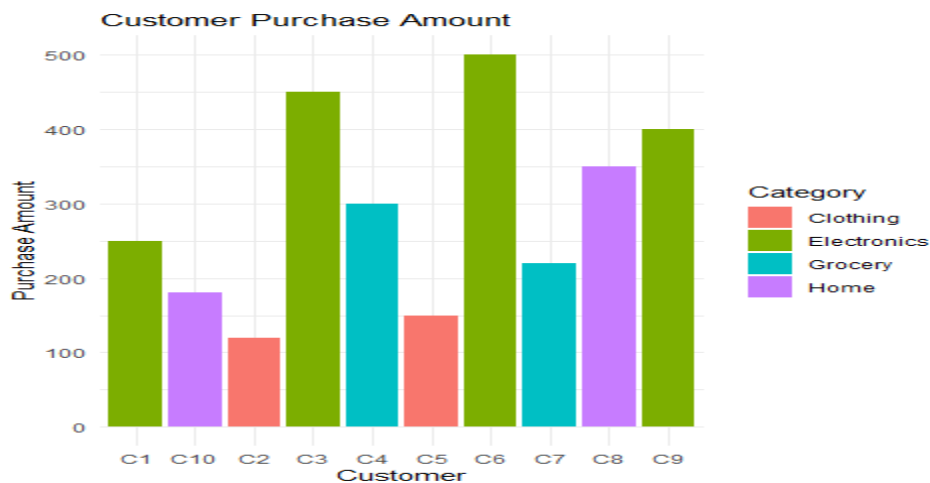
Output:



Bar chart:

```
ggplot(ecommerce, aes(x = Customer,  
  y = PurchaseAmount,  
  fill = Category)) +  
geom_bar(stat = "identity") +  
labs(title = "Customer Purchase Amount",  
  x = "Customer",  
  y = "Purchase Amount") +  
theme_minimal()
```

Output:



3. University Examination Performance Dashboard

Bar Chart:

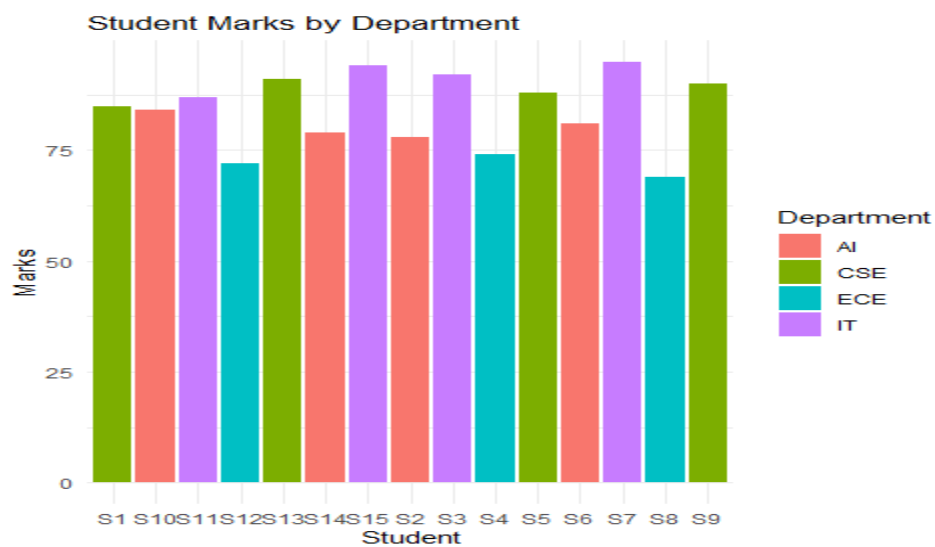
```
# Install package (Run only once)  
install.packages("ggplot2")  
  
# Load library  
library(ggplot2)  
  
# Create Dataset  
students <- data.frame(  
  Student = paste0("S", 1:15),  
  Department = c("CSE", "AI", "IT", "ECE", "CSE",  
    "AI", "IT", "ECE", "CSE", "AI",  
    "IT", "ECE", "CSE", "AI", "IT"),
```

```

Marks = c(85,78,92,74,88,
          81,95,69,90,84,
          87,72,91,79,94)
)
ggplot(students,
       aes(x = Student,
           y = Marks,
           fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Student Marks by Department",
       x = "Student",
       y = "Marks") +
  theme_minimal()

```

Output:



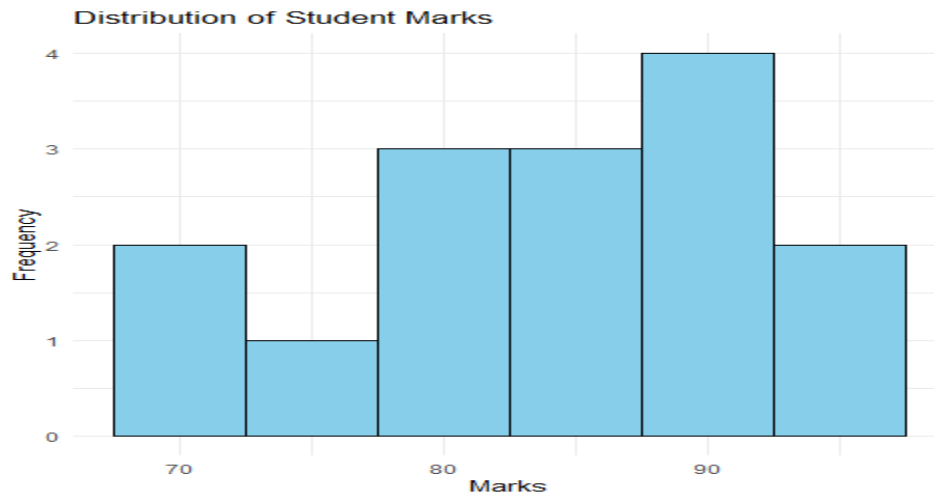
Histogram:

```

ggplot(students,
       aes(x = Marks)) +
  geom_histogram(binwidth = 5,
                fill = "skyblue",
                color = "black") +
  labs(title = "Distribution of Student Marks",

```

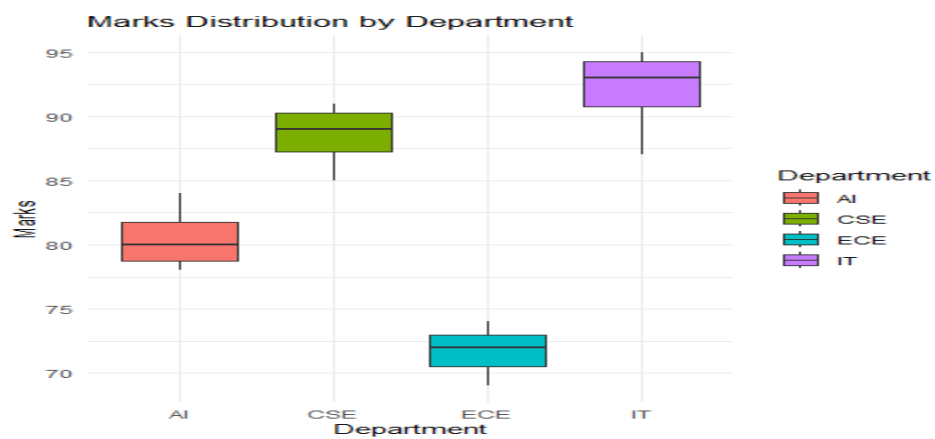
```
x = "Marks",
y = "Frequency") +
theme_minimal()
Output:
```



Box Plot:

```
ggplot(students,
aes(x = Department,
y = Marks,
fill = Department)) +
geom_boxplot() +
labs(title = "Marks Distribution by Department",
x = "Department",
y = "Marks") +
theme_minimal()
```

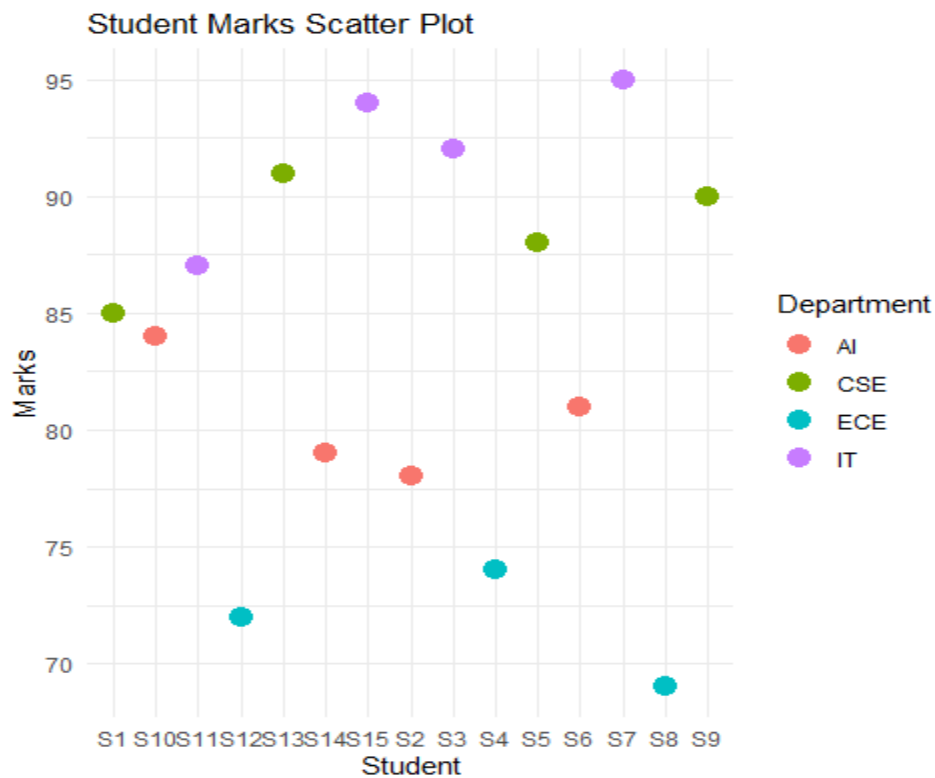
Output:



Scatter Plot:

```
ggplot(students,  
  aes(x = Student,  
    y = Marks,  
    color = Department)) +  
geom_point(size = 4) +  
labs(title = "Student Marks Scatter Plot",  
  x = "Student",  
  y = "Marks") +  
theme_minimal()
```

Output:

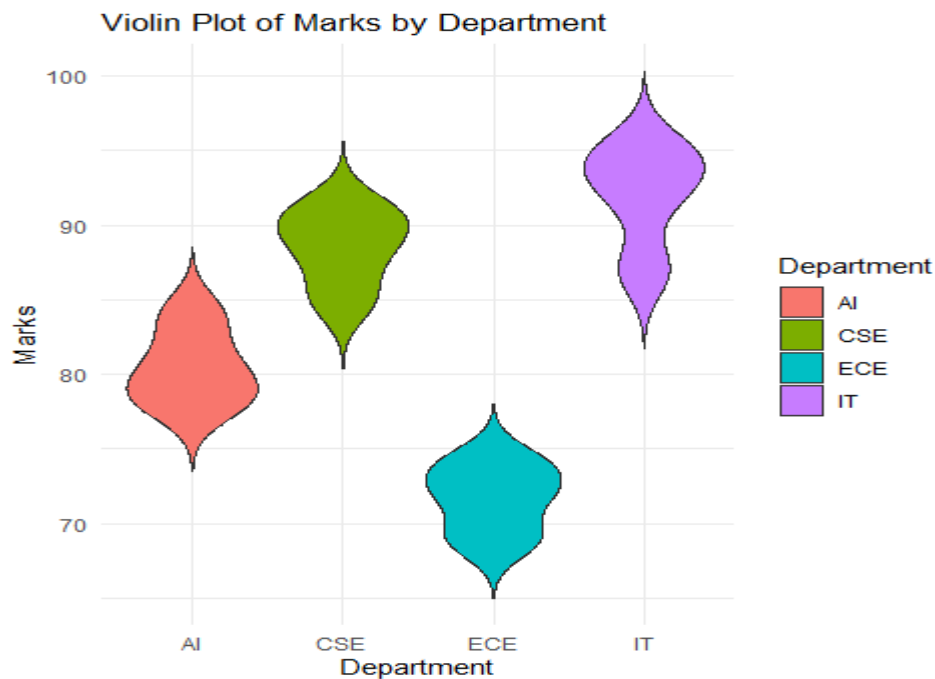


Violin Plot:

```
ggplot(students,  
  aes(x = Department,  
    y = Marks,  
    fill = Department)) +  
geom_violin(trim = FALSE) +
```

```
labs(title = "Violin Plot of Marks by Department",
     x = "Department",
     y = "Marks") +
theme_minimal()
```

Output:



4. Climate and Rainfall Distribution Study

Histogram:

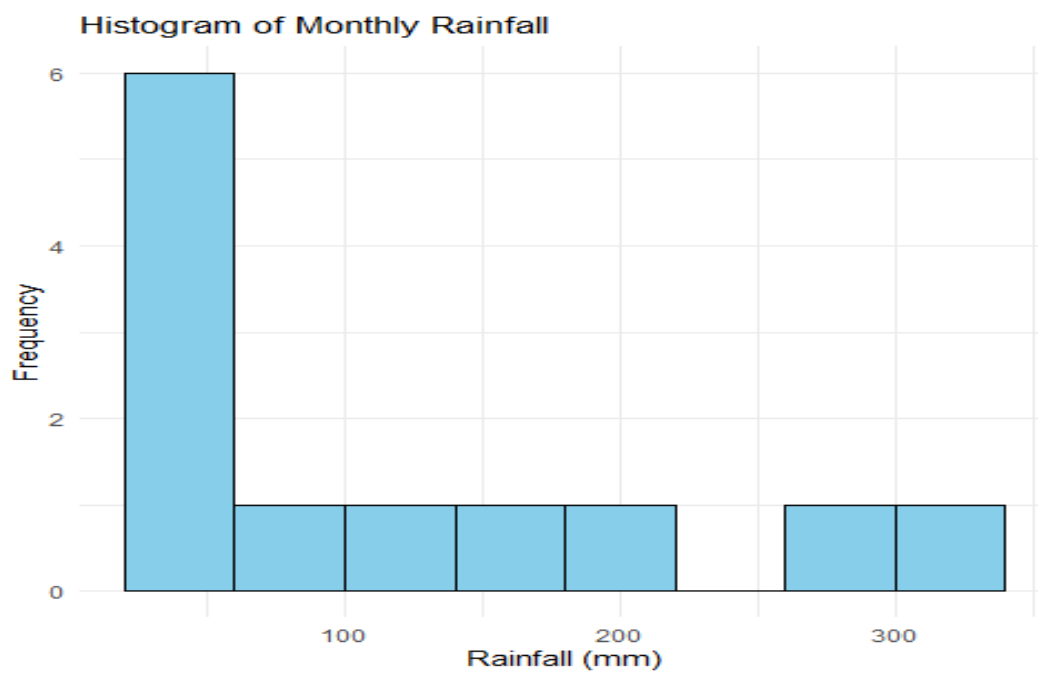
```
# Install package (Run only once)
install.packages("ggplot2")

# Load library
library(ggplot2)

# Create Dataset
climate <- data.frame(
  Month = factor(c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                  "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
                levels=c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                        "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")),
  Rainfall = c(25, 30, 40, 55, 120, 210, 320, 290, 180, 100, 60, 35),
  Temperature = c(22, 24, 27, 30, 33, 31, 29, 28, 28, 27, 25, 23),
```

```
Humidity = c(60,62,65,68,75,85,90,88,84,78,70,65)
)
ggplot(climate,
  aes(x=Rainfall))+
  geom_histogram(binwidth=40,
    fill="skyblue",
    color="black")+
  labs(title="Histogram of Monthly Rainfall",
    x="Rainfall (mm)",
    y="Frequency")+
  theme_minimal()
```

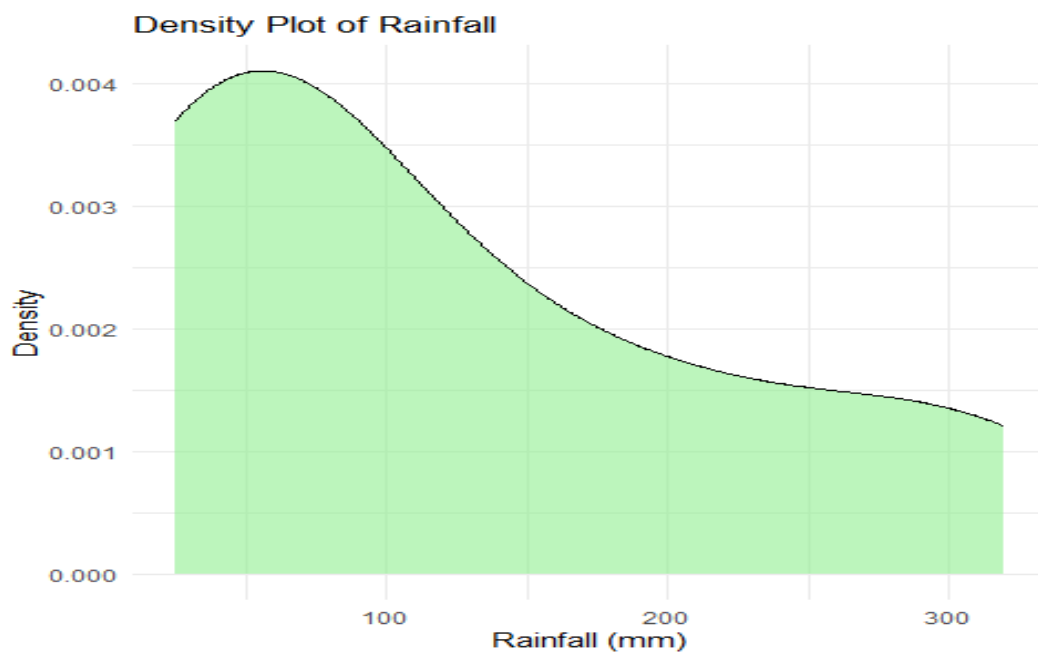
Output:



Density Plot:

```
ggplot(climate,
  aes(x=Rainfall))+
  geom_density(fill="lightgreen",
    alpha=0.6)+
  labs(title="Density Plot of Rainfall",
    x="Rainfall (mm)",
```

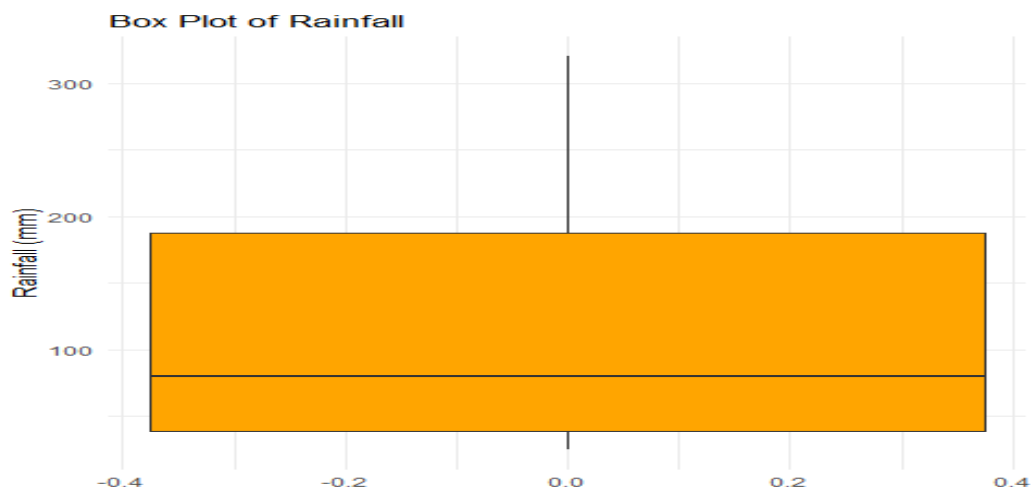
```
y="Density")+
theme_minimal()
Output:
```



Box Plot:

```
ggplot(climate,
aes(y=Rainfall))+
geom_boxplot(fill="orange")+
labs(title="Box Plot of Rainfall",
y="Rainfall (mm)")+
theme_minimal()
```

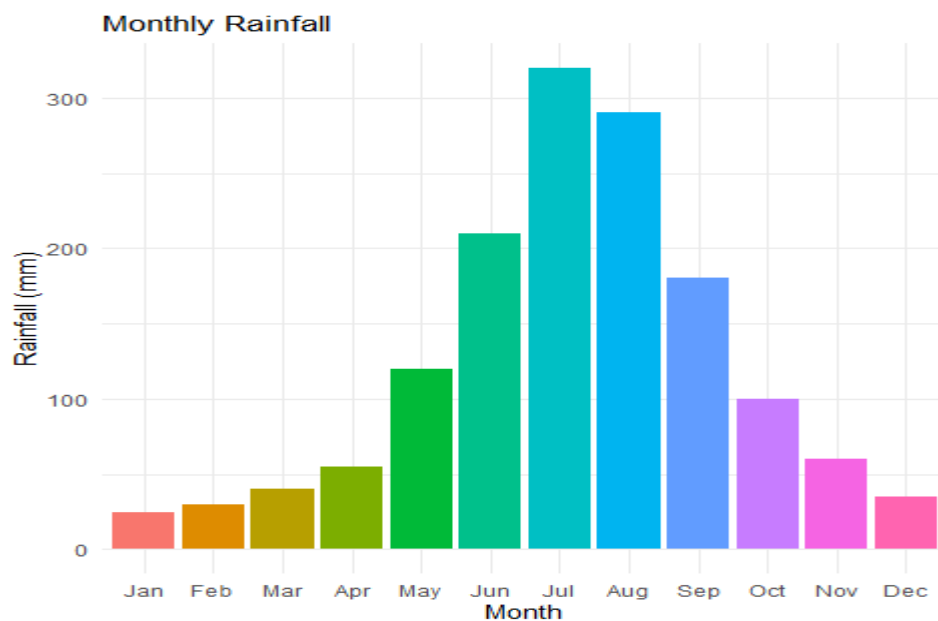
Output:



Bar Chart:

```
ggplot(climate,  
  aes(x=Month,  
    y=Rainfall,  
    fill=Month))+  
geom_bar(stat="identity")+  
labs(title="Monthly Rainfall",  
  x="Month",  
  y="Rainfall (mm)")+  
theme_minimal()+  
theme(legend.position="none")
```

Output:



5. Bank Loan Risk Assessment

Histogram:

```
# Load Library
```

```
library(ggplot2)
```

```
# Create Dataset
```

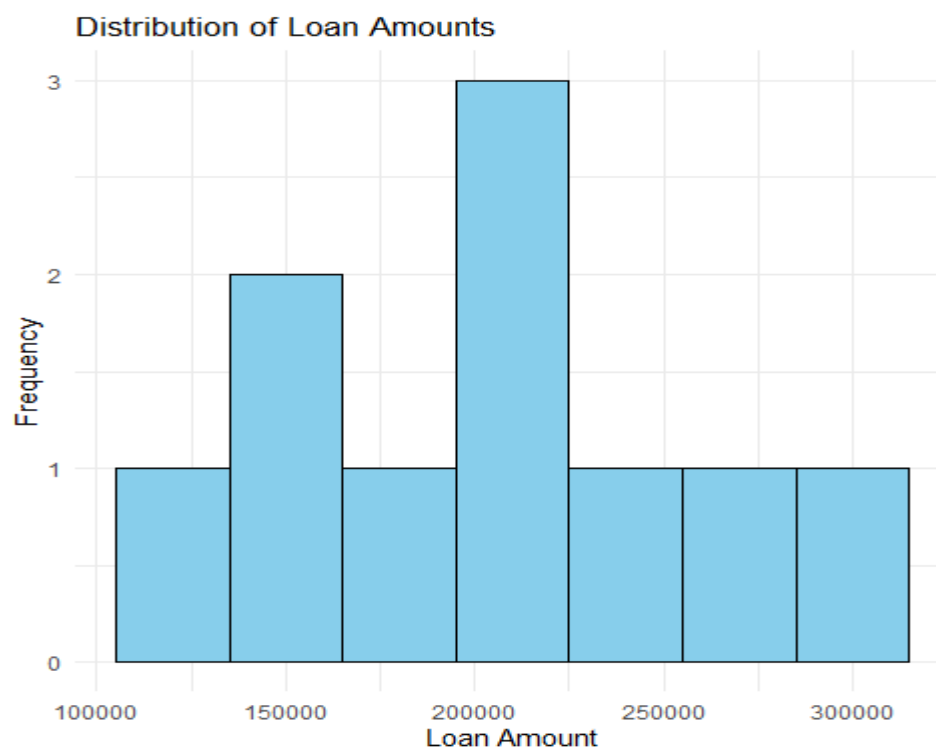
```
loan <- data.frame(  
  Customer = paste0("C", 1:10),  
  LoanAmount = c(150000, 200000, 120000, 300000, 250000,  
    180000, 220000, 160000, 280000, 210000)
```

```

      180000, 220000, 275000, 160000, 210000),
Income = c(50000, 65000, 45000, 90000, 75000,
      60000, 70000, 85000, 55000, 68000),
CreditScore = c(720, 680, 750, 620, 700,
      690, 710, 640, 730, 695),
Repayment = c("Good", "Average", "Good", "Poor", "Good",
      "Average", "Good", "Poor", "Good", "Average")
)
ggplot(loan,
      aes(x = LoanAmount)) +
geom_histogram(binwidth = 30000,
      fill = "skyblue",
      color = "black") +
labs(title = "Distribution of Loan Amounts",
      x = "Loan Amount",
      y = "Frequency") +
theme_minimal()

```

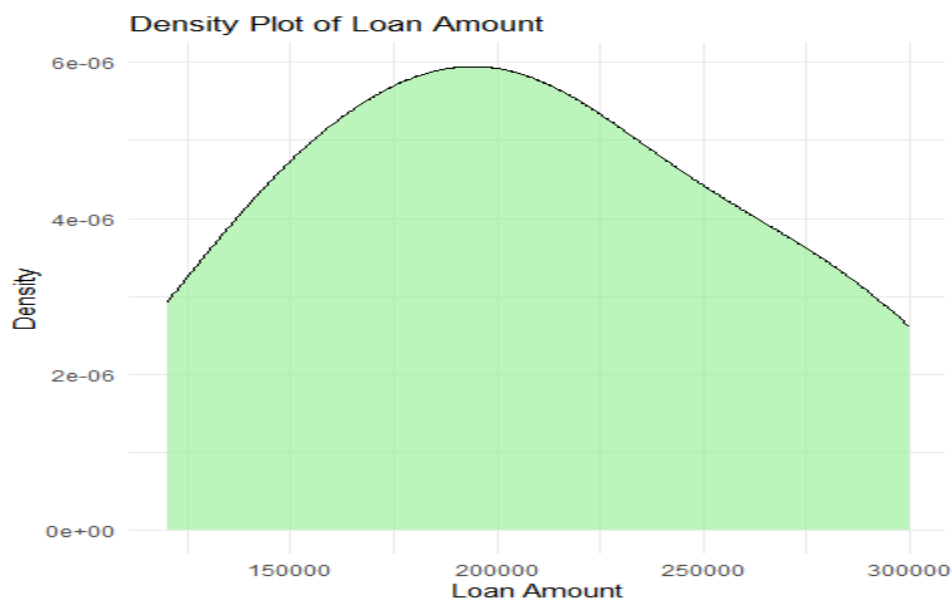
Output:



Density Plot:

```
ggplot(loan,  
  aes(x = LoanAmount)) +  
  geom_density(fill = "lightgreen",  
    alpha = 0.6) +  
  labs(title = "Density Plot of Loan Amount",  
    x = "Loan Amount",  
    y = "Density") +  
  theme_minimal()
```

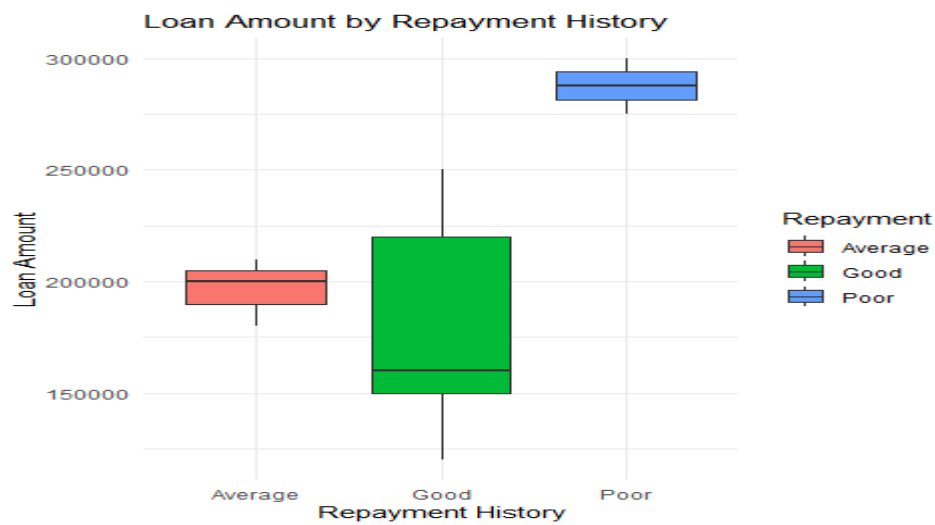
Output:



Box Plot:

```
ggplot(loan,  
  aes(x = Repayment,  
    y = LoanAmount,  
    fill = Repayment)) +  
  geom_boxplot() +  
  labs(title = "Loan Amount by Repayment History",  
    x = "Repayment History",  
    y = "Loan Amount") +  
  theme_minimal()
```

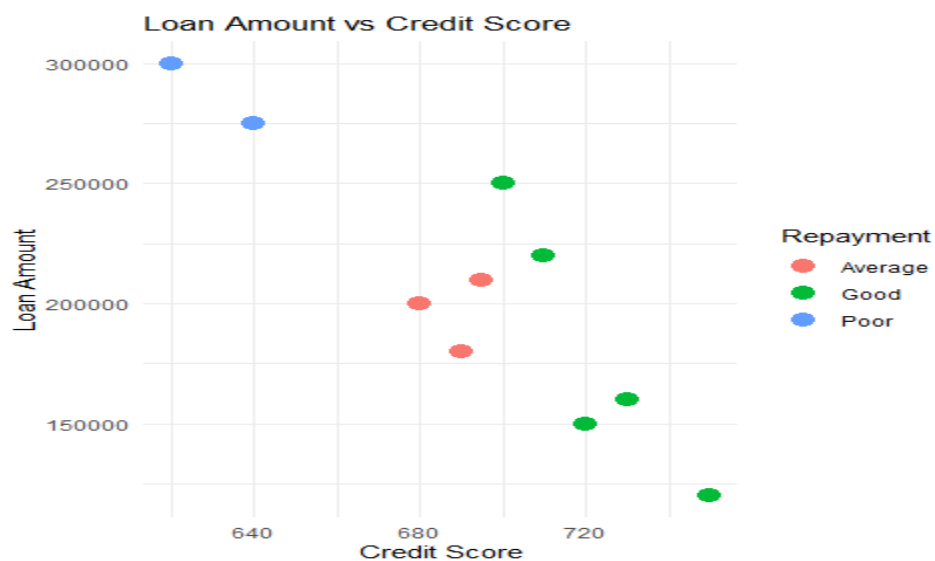
Output:



Scatter Plot:

```
ggplot(loan,
  aes(x = CreditScore,
    y = LoanAmount,
    color = Repayment)) +
  geom_point(size = 4) +
  labs(title = "Loan Amount vs Credit Score",
    x = "Credit Score",
    y = "Loan Amount") +
  theme_minimal()
```

Output:



Bar Chart:

```
ggplot(loan,  
  aes(x = Customer,  
    y = LoanAmount,  
    fill = Repayment)) +  
geom_bar(stat = "identity") +  
labs(title = "Customer Loan Amount",  
  x = "Customer",  
  y = "Loan Amount") +  
theme_minimal()
```

Output:

